

CAP-MADELEINE, QC, Canada

WMO: 714250

Lat: 49.2508N

Lon: 65.3247W

Elev: 29

StdP: 100.98

Time Zone: -5.00 (NAE)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-20.7	-18.5	-25.5	0.4	-20.3	-23.3	0.5	-18.3	18.1	-9.0	16.3	-10.2	7.6	270	0.759

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	6.3	23.8	18.3	22.0	17.2	20.6	16.4	19.2	22.7	18.1	21.1	17.1	19.9	5.1	190

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
17.8	12.8	21.4	16.7	11.9	20.0	15.8	11.2	19.0	55.2	22.7	51.4	21.2	48.3	20.0	23.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
15.2	13.3	11.9	-23.3	28.6	2.6	1.9	-25.1	30.0	-26.6	31.2	-28.1	32.3	-29.9	33.7		
			DB	WB	DB	WB	DB	WB	DB	WB	DB	WB	DB	WB		
			-23.7	21.5	2.6	1.0	-25.6	22.2	-27.1	22.8	-28.6	23.3	-30.5	24.0		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	4.1	-9.0	-8.7	-4.2	1.8	7.5	13.0	16.7	16.2	12.2	6.7	1.2	-4.5
	DBStd	9.78	5.39	5.34	4.92	3.53	3.37	3.39	2.75	2.81	3.33	3.43	4.09	4.45
	HDD10.0	2746	588	525	441	245	94	9	0	0	14	114	264	450
	HDD18.3	5211	847	758	699	494	335	163	67	79	186	360	513	708
	CDD10.0	609	0	0	0	1	17	100	207	191	79	13	2	0
	CDD18.3	31	0	0	0	0	0	3	15	12	1	0	0	0
	CDH23.3	78	0	0	0	1	1	11	35	27	4	0	0	0
	CDH26.7	10	0	0	0	0	0	2	4	3	1	0	0	0
Wind	WSAvg	5.5	7.0	6.5	6.2	5.8	4.8	4.4	4.1	4.1	4.7	5.5	6.1	6.9
Precipitation	PrecAvg	897	72	58	75	64	61	68	85	86	67	72	64	105
	PrecMax	1203	169	201	209	155	145	120	208	168	157	160	182	218
	PrecMin	691	24	19	29	20	17	28	27	39	24	15	19	31
	PrecStd	154	38	37	39	35	28	24	40	31	34	41	33	45
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	8.6	6.9	9.3	15.2	21.0	24.9	26.7	26.0	23.2	19.7	16.9	8.9
		MCWB	7.1	5.0	5.6	10.8	13.6	18.5	19.3	19.4	18.3	14.3	14.1	6.4
	2%	DB	3.2	2.2	5.8	10.5	17.0	21.7	23.8	23.5	20.7	16.3	11.7	5.1
		MCWB	1.5	0.8	3.0	6.4	11.7	16.1	18.6	18.3	16.3	12.5	8.8	3.5
	5%	DB	0.5	0.2	3.6	7.9	14.2	19.4	22.0	21.6	18.7	13.4	8.4	3.2
		MCWB	-0.4	-0.8	1.2	4.7	9.8	14.7	17.5	17.3	15.1	10.6	6.0	1.7
	10%	DB	-1.5	-1.3	1.7	6.1	12.2	17.7	20.5	20.1	17.0	11.5	6.3	1.3
		MCWB	-2.4	-2.1	-0.1	3.4	8.6	13.7	16.7	16.5	13.8	8.9	4.3	0.2
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	7.2	5.3	6.7	11.3	15.2	19.7	21.1	20.6	19.1	15.4	13.8	6.9
		MCDB	9.1	7.1	9.1	14.9	19.3	23.6	24.9	24.9	22.2	18.6	16.9	8.7
	2%	WB	1.6	1.2	3.3	6.7	12.5	17.3	19.5	19.0	17.1	12.7	8.7	3.7
		MCDB	2.8	2.0	5.2	9.5	16.0	20.6	22.7	22.5	19.8	15.8	11.1	4.7
	5%	WB	-0.6	-0.6	1.5	5.0	10.5	15.5	18.2	17.9	15.5	10.7	6.1	1.8
		MCDB	0.3	0.1	3.4	7.5	13.6	18.5	21.1	20.8	18.1	13.2	7.8	3.0
	10%	WB	-2.4	-1.9	0.2	3.7	9.1	14.3	17.2	16.9	14.1	9.1	4.6	0.3
		MCDB	-1.7	-1.2	1.5	5.7	11.8	17.2	20.0	19.6	16.6	11.1	5.9	1.1
Mean Daily Temperature Range	5% DB	MDBR	6.8	6.7	6.3	5.8	6.5	6.9	6.3	6.0	6.2	5.3	5.1	5.3
		MCDBR	10.5	8.1	8.4	9.3	10.3	9.6	8.6	8.3	9.2	9.0	8.7	8.2
		MCWBR	10.1	7.3	6.3	6.5	6.5	5.7	5.1	5.2	6.3	6.8	7.5	7.5
	5% WB	MCDBR	10.5	7.4	7.7	8.7	9.7	8.7	7.9	7.6	8.6	8.8	8.4	7.7
		MCWBR	10.0	6.8	6.2	6.5	6.6	6.0	5.3	5.2	6.3	6.9	7.4	7.1
Clear-Sky Solar Irradiance	taub	0.267	0.276	0.301	0.369	0.362	0.380	0.416	0.380	0.357	0.336	0.296	0.272	
	taud	2.372	2.307	2.313	2.255	2.413	2.377	2.326	2.423	2.459	2.495	2.515	2.416	
	Ebn at Noon	800	879	912	876	897	879	840	857	844	802	767	745	
	Edh at Noon	65	88	104	124	111	116	121	105	91	75	58	55	
All-Sky Solar Radiation	RadAvg	1.00	1.95	3.21	4.39	5.15	5.68	5.44	4.72	3.55	2.17	1.09	0.74	
	RadStd	0.10	0.17	0.27	0.35	0.33	0.47	0.49	0.25	0.26	0.15	0.12	0.08	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (16 neighbors)	N/A	N/A	+1.32	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page